

Representational drift as a result of implicit regularization

Aviv Ratzon , Dori Derdikman, Omri Barak

Rappaport Faculty of Medicine, Technion - Israel Institute of Technology, Haifa 31096, Israel • Network Biology Research Laboratory, Technion - Israel Institute of Technology, Haifa 32000, Israel

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

October 10, 2023 (this version)

Sent for peer review

June 23, 2023

Posted to bioRxiv

June 21, 2023

Abstract

Recent studies show that, even in constant environments, the tuning of single neurons changes over time in a variety of brain regions. This representational drift has been suggested to be a consequence of continuous learning under noise, but its properties are still not fully understood. To uncover the underlying mechanism, we trained an artificial network on a simplified navigational task, inspired by the predictive coding literature. The network quickly reached a state of high performance, and many neurons exhibited spatial tuning. We then continued training the network and noticed that the activity became sparser with time. We observed vastly different time scales between the initial learning and the ensuing sparsification. We verified the generality of this phenomenon across tasks, learning algorithms, and parameters. This sparseness is a manifestation of the movement within the solution space - the networks drift until they reach a flat loss landscape. This is consistent with recent experimental results demonstrating that CA1 neurons increase sparseness with exposure to the same environment and become more spatially informative. We conclude that learning is divided into three overlapping phases: Fast familiarity with the environment, slow implicit regularization, and a steady state of null drift. The variability in drift dynamics opens the possibility of inferring learning algorithms from observations of drift statistics.

eLife assessment

This study presents a new and **valuable** theoretical account of spatial representational drift in the hippocampus. The evidence supporting the claims is **convincing**, with a clear and accessible explanation of the phenomenon. Overall, this study will likely attract researchers exploring learning and representation in both biological and artificial neural networks.

What do we mean when we say that the brain represents the external world? One interpretation is the existence of neurons whose activity is tuned to world variables. Such neurons have been observed in many contexts: place cells [1, 2] – which are tuned to position in a specific context, visual cells [3] – which are tuned to specific visual cues, neurons that are tuned to the execution of actions [4] and more. This tight link between the external world and neural activity might suggest that, in the absence of environmental or behavioral changes, neural activity is constant. In contrast, recent studies show that, even in constant environments, the tuning of single neurons to outside world variables gradually changes over time in a variety of brain regions, even long after good representations of the stimuli were achieved. This phenomenon has

been termed *representational drift*, and has changed the way we think about the stability of memory and perception, but its driving forces and properties are still unknown [5, 6, 7, 8] (see [9, 10] for an alternative account).

There are at least two immediate theoretical questions arising from the observation of drift – why does it happen, and whether and how behavior is resistant to it [11, 12]? One mechanistic explanation is that the underlying anatomical substrates are themselves undergoing constant change, such that drift is a direct manifestation of this structural morphing [13]. A normative interpretation posits that drift is a solution to a computational demand, such as temporal encoding [14], ‘drop-out’ regularization [15], exploration of the solution space [16], or re-encoding during continual learning [11]. Several studies also address the resistance question, providing possible explanations on how behavior can be robust to such phenomena [17, 18, 19, 20].

Here, we focus on the mechanistic question, and leverage analyses of drift statistics for this purpose. Specifically, recent studies showed that representational drift in the CA1 is driven by active experience [21, 22]. Namely, rate maps decorrelate more when mice are active for a longer time in a given context. This implies that drift is not just a passive process, but rather an active learning one. As drift seems to occur after an adequate representation has formed, it seems fitting to model it as a form of a continuous learning process.

This approach has been recently explored by [23, 24]. They considered continuous learning in noisy, overparameterized neural networks. Because the system is overparameterized, a manifold of zero-loss solutions exists. [23] showed that for feedforward neural networks (FNNs) trained using Hebbian learning with added parameter noise, neurons change their tuning over time. This was due to an *undirected* random walk within the manifold of solutions. The coordinated drift of neighboring place fields was used as evidence to support this view. The phenomenon of undirected motion within the space of solutions seems plausible, as all members of this space achieve equally good performance (**Fig 1A** left). However, there may be other properties of the solutions (**Fig 1B**) that vary along this manifold, which could potentially bias drift in a certain direction (**Fig 1A** right). It is likely that the drift observed in experiments is a combination of both an undirected and directed movement. We will now introduce theoretical results from machine learning that support the possibility of directed drift.

Recent work provided a tractable analytical framework for the learning dynamics of Stochastic Gradient Descent (SGD) with added noise and an overparameterized regime [25, 26]. These studies showed that, after the network has converged to the zero-loss manifold, a second-order effect biases the random walk along a specific direction within this manifold. This direction reduces an implicit regularizer, determined by the type of noise the network is exposed to. The regularizer is related to the Hessian of the loss – a measure of the flatness of the loss landscape in the vicinity of the solutions. Since this directed movement is a second-order effect, its timescale is orders of magnitude larger than that of the initial convergence.

Consider a biological neural network performing a task. The ML implicit regularization mentioned above requires three components: an overparameterized regime, noise, and SGD. Both biological and artificial networks possess a large number of synapses, or parameters, and hence can reasonably be expected to be overparameterized. Noise can emerge from the external environment or from internal biological elements. It is not reasonable to assume that a precise form of gradient descent is implemented in the brain [27], thereby casting doubt on the third element. Nevertheless, biologically plausible rules could be considered as noisy versions of gradient descent, as long as there is a coherent improvement in performance [28, 29]. Motivated by this analogy, we explore representational drift in models and experimental data.

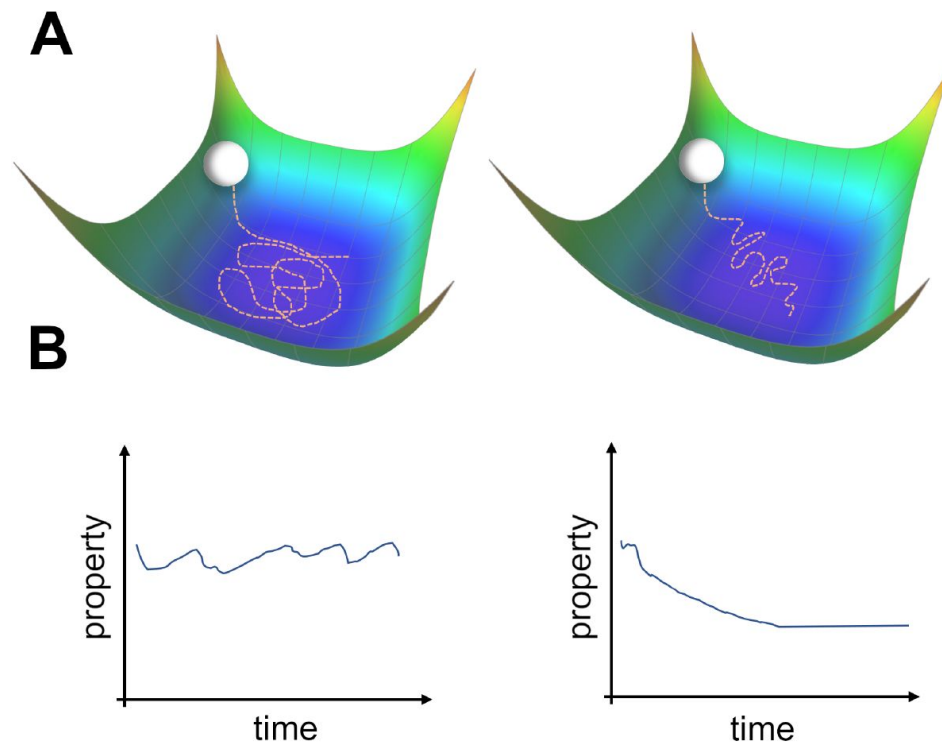


Figure 1

Two types of possible movements within the solution space.

(A) Two options of how drift may look in the solution space. Random walk within the space of equally good solutions that is either undirected (left) or directed (right). (B) The qualitative consequence of the two movement types. For an undirected random walk, all properties of the solution will remain roughly constant (left). For the directed movement there should be a given property that is gradually increasing or decreasing (right).

Because drift is commonly observed in spatially-selective cells, we base our analysis on a model which has been shown to contain such cells [30]. Specifically, we trained artificial neural networks on a predictive coding task in the presence of noise. In this task, an agent moves along a linear track while receiving visual input from the walls, such that the goal is to predict the subsequent input. We observed that neurons became tuned to the latent variable, which is position, in accordance with previous results [30]. We continued training and found that in addition to the gradual change of tuning curves, similar to [23], we witnessed that the number of active neurons decreased slowly while their tuning specificity increased. These results align with recent experimental observations [21]. Finally, we demonstrated the connection between this sparsification effect and changes to the Hessian, in accordance with ML theory.

Results

Spontaneous sparsification in a predictive coding network

To model representational drift in the CA1 area, we chose a simple model that could give rise to spatially-tuned cells [30]. In this model, an agent traverses a corridor while slightly modulating its angle with respect to the main axis (Fig 2A). The walls are textured by a fixed smooth noisy signal, and the agent receives this as input according to its current field of view. The model itself is a single hidden layer feedforward network, with the velocity and visual field as inputs. The desired output is the predicted visual input in the next time step. The model equations are given by:

$$\hat{\mathbf{y}}_t = \sigma(\mathbf{x}_t \mathbf{m}^T + \mathbf{b}) \mathbf{n}^T, \quad (1)$$

where $\mathbf{m}$ and $\mathbf{n}$ are the input and output matrices respectively, $\mathbf{b}$ is the bias vector, and σ is the ReLU activation function. The task is for the network's output, $\mathbf{y}$, to match the visual input, $\mathbf{x}$ of the following time step, resulting in the following loss function:

$$f(\mathbf{m}, \mathbf{n}, \mathbf{b}) = \mathbb{E}_t (\hat{\mathbf{y}}_t - \mathbf{x}_{t+1})^2. \quad (2)$$

We train the network using Gradient Descent (GD), while adding update noise to the learning dynamics:

$$\theta_{\tau+1} = \theta_{\tau} - \eta \frac{\partial f(\theta_{\tau})}{\partial \theta_{\tau}} + \xi_{\tau}^{update}, \quad (3)$$

where $\theta = (\mathbf{m}, \mathbf{n}, \mathbf{b})$ is the vectorized parameters-vector, τ is the current training step and ξ_{τ}^{update} is

Gaussian noise. We let the network converge to a good solution, demonstrated by a loss plateau, and continue training for an additional period. Note that this additional period can be orders of magnitude longer than the initial training period. The network quickly converged to a low loss and stayed at the same loss during the additional training period (Fig 2B). Surprisingly, when looking at the activity within the hidden layer, we noticed that it slowly became sparse. This sparsification did not hurt performance, because individual units became more informative, as quantified by the average mutual information between unit activity and the position of the agent (Fig 2C). When looking at the rate maps of neurons, i.e. their tuning to position, one can observe an image similar to representational drift observed in experiments [5] – namely that neurons changed their tuning over time (Fig 2D). Additionally, their tuning specificity increased in accordance with the information increase. By observing the correlation matrix of the rate maps over time, it is apparent that there was a gradual change that slowed down (Fig 2E). To summarize, we observed a spontaneous sparsification over a timescale much longer than the initial convergence, without introducing any explicit regularization. This is comparable to experimental data from [21], where indeed drift was characterized by a decrease in the fraction

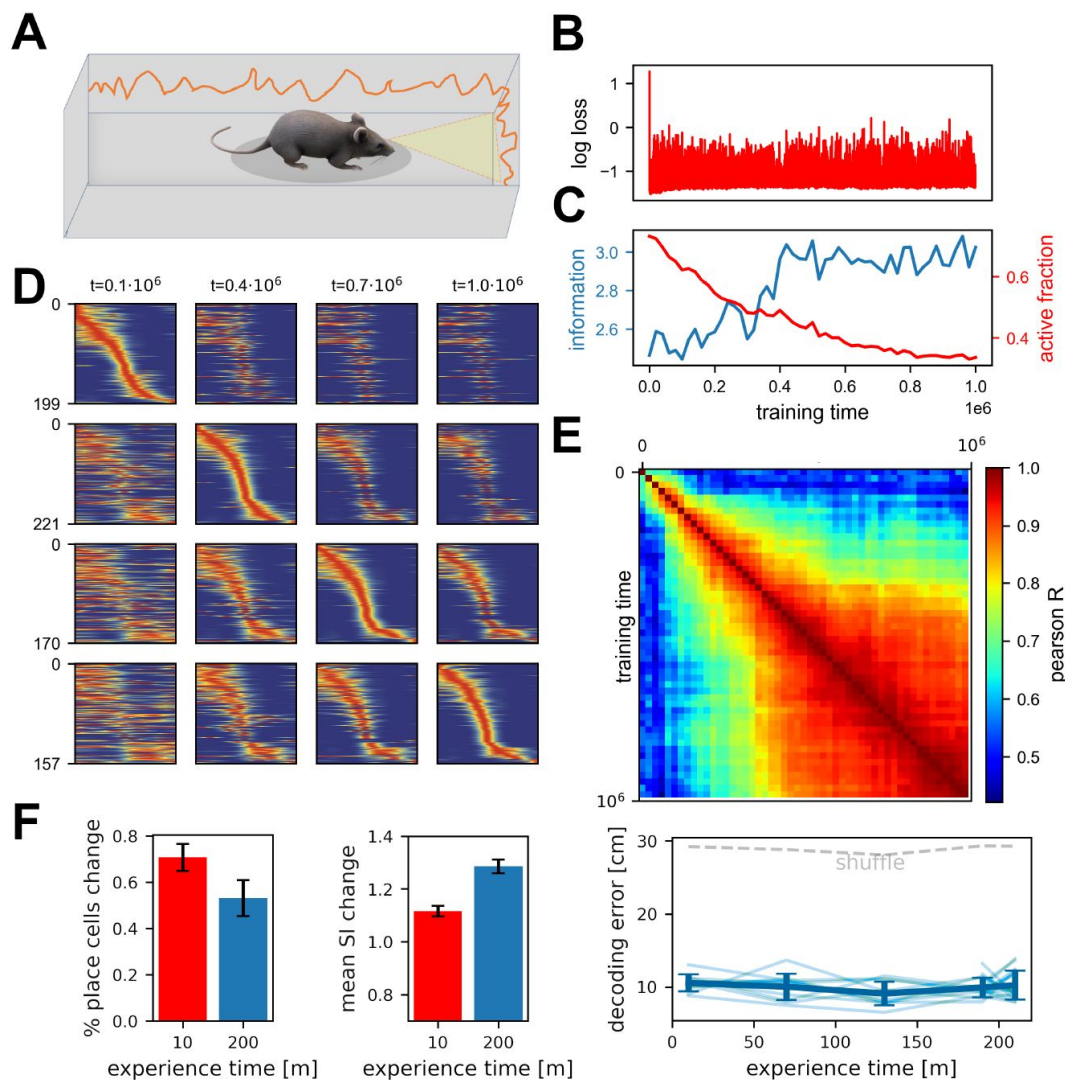


Figure 2

Noisy learning leads to spontaneous sparsification.

(A) Illustration of an agent in a corridor receiving high-dimensional visual input from the walls. (B) Log loss as a function of training steps, log loss of 0 corresponds to a mean estimator. The loss rapidly decreases, and then remains roughly constant. (C) Information (blue) and fraction of units with non-zero activation for at least one input (red) as a function of training steps. (D) Rate maps sampled at four different time points. Maps in each row are sorted according to a different time point. Sorting is done based on the peak tuning value to the latent variable. (E) Correlation of rate maps between different time points along training. Only active units are used. (F) Figures reproduced from [21] where mice spent different amount of time in two environments. Fraction of place cells in the beginning relative to the end of the experiment (left), average Spatial Information (SI) per cell in the beginning relative to the end of the experiment (middle) and the decoding error for the position of the mouse (right).

of active place cells, and an increase in cells' information while the decoding error for the position of the mouse stayed relatively constant (**Fig 2F**). [31] also reported a decrease in CA1 neural activity and rise in specificity with environment familiarity. Another recent study further demonstrated an increase in information over days [32].

Generality of the phenomenon

To explore the sensitivity of our results to specific modeling choices, we systematically varied many of them (**Fig 3A**). Specifically, we replaced the task with either a simplified predictive coding, random mappings or smoothed random mappings. Noise was introduced to the outputs (label noise), instead of the update noise. We simulated different activation functions. Perhaps most important, we varied the learning rules, as SGD is not a biologically plausible one. We used both Adam [33] and RMSprop [34], from the ML literature. We also used Stochastic Error-Descent (SED) [35], which does not require gradient calculation and is more biologically plausible (5). All cases demonstrated an initial, fast, phase of convergence to low loss, followed by a much slower phase of directed random motion within the low-loss space.

The results of the simulations supported our main conclusion, though several qualitative phenomena could be observed. First of all, sparsification dynamics were not sensitive to most of the parameters. The main qualitative difference observed was that the timescales could vary by orders of magnitude as a function of the noise scale (**Fig 3B** bottom). Note that we calculate the timescale of sparsification by fitting an exponential curve to the fraction of active units over time, and take the time constant of the fitted exponential. Additionally, apart from simulations that did not converge due to too big timescales, the final sparsity was the same for all networks of the same size (**Fig 3B** top), in accordance with results from [23]. In a sense, once noise is introduced the network is driven to maximal sparsification. For Adam, RMSprop and SED sparsification ensued in the absence of any added noise. For SED the explanation is straightforward, as the parameter updates are driven by noise. For Adam and RMSprop, we suggest that in the vicinity of the zero-loss manifold, the second moment acts as noise.

For label noise, the dynamics were qualitatively different, the fraction of active units did not reduce, but the activity of the units did sparsify. In some cases, the networks quickly collapsed to a sparse solution, most likely as a result of the learning rate being too high, in relation to the input statistics [36]. Importantly, for GD without noise, there was no change after the initial convergence.

As a further test of the generality of this phenomenon, we consider the recent simulation from [23]. The learning rule used in this work was very different from the ones we applied. We, therefore, simulated that network using the published code. We found the same type of dynamics as shown above, namely that the network initially converged to a good solution followed by a longer period of sparsification (**Fig 3C**). Note that in The original publication [23] the focus was on the stage following this sparsification, in which the network indeed maintained a constant fraction of active cells.

In conclusion, we see that noisy learning leads to three phases under rather general conditions. First, fast learning of the task and convergence to the manifold of low-loss solutions. The second phase is directed movement on this manifold driven by a second-order effect of implicit regularization. The third phase is an undirected random walk within the sub-manifold of low loss and maximum regularization.

Mechanism of sparsification

What are the mechanisms that give rise to sparsification? As illustrated in **Fig. 1**, different solutions in the zero-loss manifold might vary in some of their properties. The specific property suggested from theory [25] is the flatness of the loss landscape in the vicinity of the solution.

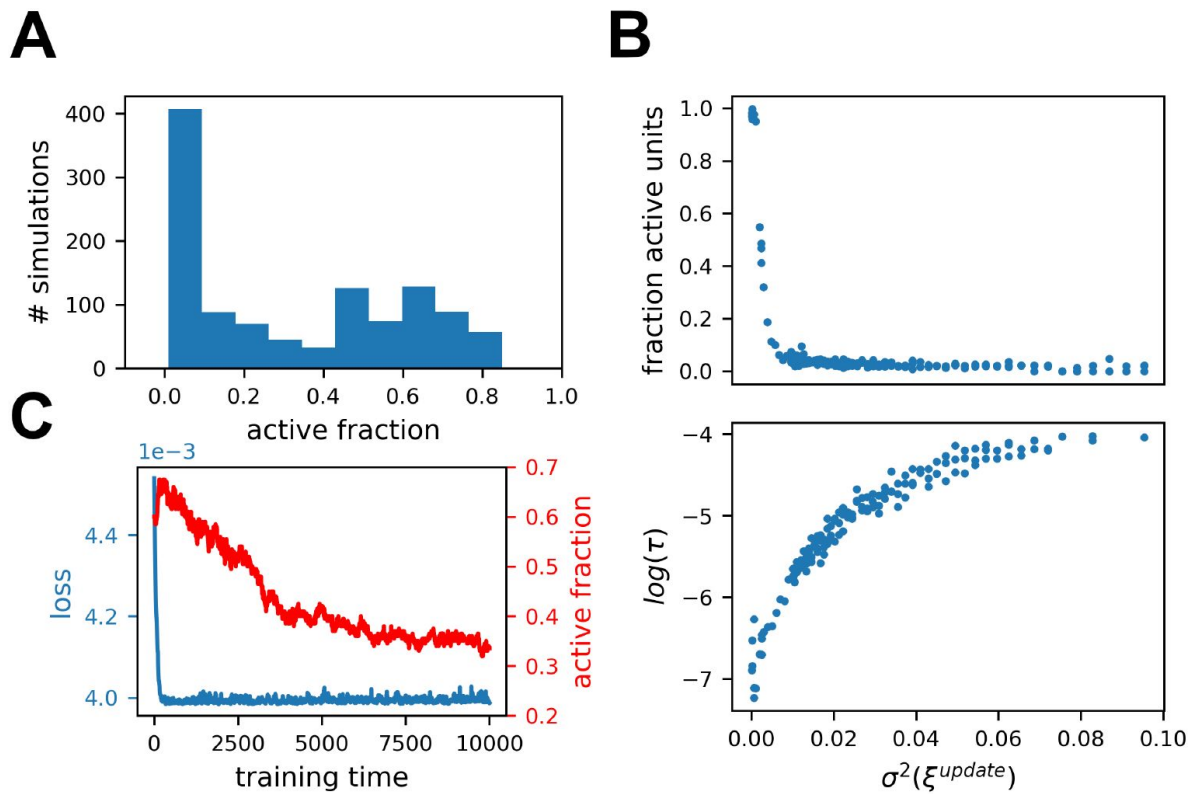


Figure 3

Generality of the results.

Summary of 1117 simulations with various parameters (see [Table 1](#)). (A) Histogram of fraction of active units after 10^7 training steps for each simulation. (B) Subset of 178 simulations with the same parameters and varying noise variance, each point represents a single simulation. Fraction of active units as a function of the variance of the noise (top), the log of sparsification time scale as a function of the variance of the noise (bottom). (C) Learning a similarity matching task with Hebbian and anti-Hebbian learning with published code from [\[23\]](#). Performance of the network (blue) and fraction of active units (red) as a function of training steps. Note that the loss axis does not start at zero, and the dynamic range is small.

Parameter	Possible values
learning algorithm	{SGD, Adam, SED}
noise type	{update, label}
number of samples	[1,20]
initialization regime	{lazy, rich}
task	{abstract predictive, random, random smoothed}
input dimension	[1,20]
output dimension	[1,20]
noise variance (label/update)	[0.1,1]/[0.01,0.1]
hidden layer size	100

Table 1

Parameter ranges for random simulations.

This can be demonstrated with a simple example. Consider a two-dimensional loss function. The function is shaped like a valley with a continuous one-dimensional zero-loss manifold at its bottom (**Fig 4A** [↗](#)). Crucially, the loss on the entire manifold is exactly zero, while the vicinity of the manifold becomes systematically flatter in one direction. We simulated gradient descent with added noise on this function from a random starting point (red dot). The trajectory quickly converged to the zero-loss manifold, and began a random walk on it. This walk was clearly biased towards the flatter area of the manifold, as can be seen by the spread of the trajectory. This bias could be comprehended by noting that the gradient was orthogonal to the contour lines of the loss, and therefore had a component directed towards the flat region.

In higher dimensions, flatness is captured by the eigenvalues of the Hessian of the loss. Because these eigenvalues are a collection of numbers, different scenarios could lead to minimizing different aspects of this collection. Specifically, according to [\[25\]](#), update noise should regularize the sum of the log of the non-zero eigenvalues while label noise should do the same for the sum of eigenvalues. In our predictive coding example, where update noise was added, each inactivated unit translates into a set of zero-rows in the Hessian, and thus also into a set of zero-eigenvalues (**Fig 4B** [↗](#)). The slope of the regularizer approaches infinity as the eigenvalue approaches zero, and thus small eigenvalues are driven to zero much faster than large eigenvalues (**Fig 4C** [↗](#)). So in this case, update noise leads to an increase in the number of zero eigenvalues, which are manifested as a sparse solution. Another, perhaps more intuitive, way to understand these results is that units below the activation threshold are insensitive to noise perturbations. In other scenarios, in which we simulated with label noise, we indeed observed a gradual decrease in the sum of eigenvalues (**Fig 4D** [↗](#)).

Discussion

We showed that representational drift could arise from ongoing learning in the presence of noise, after a network has already reached good performance. We suggest that learning is divided into three overlapping phases: a fast initial phase, where good performance is achieved, a second slower phase in which *directed* drift along the low-loss manifold leads to an implicit regularization and finally, a third *undirected* phase ensues once the regularizer is minimized. In our results, the directed component was associated with sparsification of the neural code, a phenomenon we also observed in experimental data.

Interpreting drift as a learning process has recently been suggested by [\[23\]](#), [\[24\]](#). Both studies focused on the final phase in which the statistics of the representations were constant. Experimentally, [\[7\]](#) reported a decrease in activity at the beginning of the experiment, which they suggested was correlated with some behavioral change, but we believe it could also be a result of the directed drift phase. [\[37\]](#) also reported a slow directed change in representation long after familiarity with the stimuli. There is another consequence of the timescale separation. Unlike in the setting of drift experiments, natural environments are never truly constant. Thus, it is possible that the second phase of learning never stops because the task is slowly changing. This would imply that the second, directed, phase may be the natural regime in which neural networks reside.

Here, we reported directed drift in the space of solutions of neural networks. This drift could be observed by examining changes to the representation of external world variables, and hence is related to the phenomenon of representational drift. Note, however, that representations are not a full description of a network's behavior [\[38\]](#). The statistics of representational changes can be used as a window into changes of network dynamics and function.

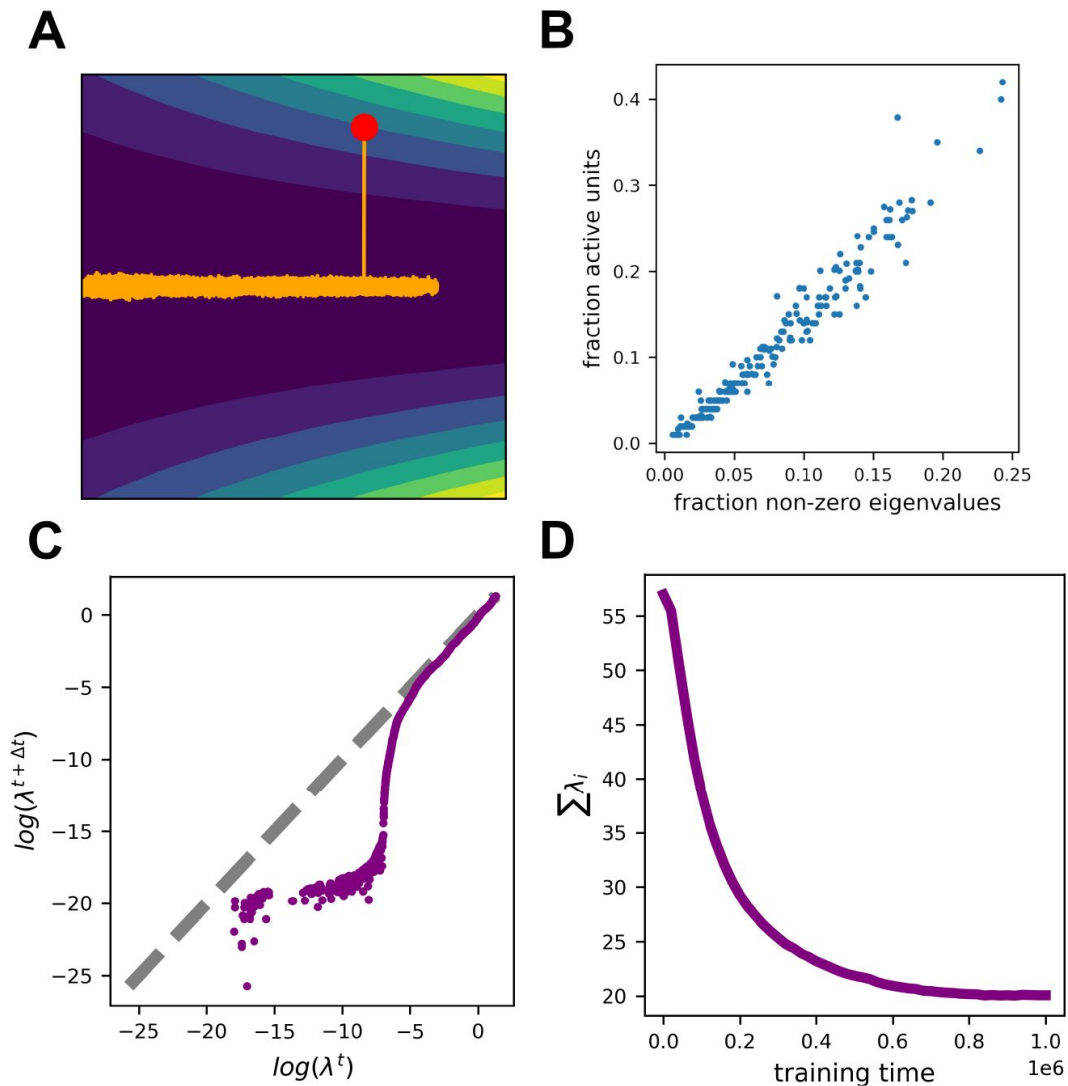


Figure 4

Noisy learning leads to a flat landscape.

(A) Gradient Descent dynamics over a two-dimensional loss function with a one-dimensional zero-loss manifold. Note that the loss is identically zero along the horizontal axis, but the left area is flatter. The orange trajectory begins at the red dot. Note the asymmetric extension into the left area. (B) Fraction of active units as a function of the number of non-zero eigenvalues. (C) Log of non-zero eigenvalues at two consecutive time points. Note that eigenvalues do not correspond to one another when calculated at two different time points, and this plot demonstrates the change in their distribution rather than changes in eigenvalues corresponding to specific directions. The distribution of larger eigenvalues hardly changes, while the distribution of smaller eigenvalues is pushed to smaller values. (D) Sum of the Hessian's eigenvalues as a function of time for learning with label noise.

The phenomenon of directed drift is very robust to various modeling choices, and also consistent with recent theoretical results [25, 26]. The details of the direction of the drift, however, are dependent on specific choices. Specifically, which aspects of the Hessian are minimized during the second phase of learning, as well as the timescale of this phase, depend on the specifics of the learning rule and the noise in the system. This suggests an exciting opportunity – inferring the learning rule of a network from the statistics of representational drift.

Our explanation of drift invoked the concept of a low-loss manifold – a family of network configurations that have identical performance on a task. The definition of low-loss, however, depends on the specific task and context analyzed. Challenging a system with new inputs could dissociate two configurations that otherwise appear identical [39]. It will be interesting to explore whether various environmental perturbations could uncover the motion along the low-loss manifold in the CA1 population. For instance, remapping was interpreted as an indicator of the detection of a context switch [40]. One can therefore speculate that the probability for remapping given the same environmental change will systematically vary as the network moves to flatter areas of the loss landscape.

Machine learning has been suggested as a model tool for neuroscience research [41, 42, 43]. However, the implicit regularization in ML has not been studied to explain representational drift in neuroscience, and may have been done without awareness of this phenomenon. It's worth noting that this isn't a phenomenon specific to neural networks, but rather a general property of overparameterized systems that optimize a cost function. Importing insights from this domain into neuroscience shows the utility of studying general phenomena in systems that learn. For example, another complex learning system in which a similar idea has been proposed is evolution – “survival of the flattest” suggests that, under a high mutation rate, the fittest replicators are not just the ones with the highest fitness, but also with a flat fitness function which is more robust to mutations [44]. One can hope that more such insights will arise as we open our eyes.

Materials and methods

Predictive coding task

The agent is moving in an arena of size (L_x, L_y) , with constant velocity in the y direction of V_0 . The agent's heading direction is θ and it changes at every time step by $\Delta\theta \sim G(0, \sigma_\theta^2)$, the agent's visual field has an angle θ_{vis} and is represented as a vector of size L_{vis} . The texture of the walls is generated from a random Gaussian vector of size $L_{walls} = 2(L_x + L_y)L_{vis}$, smoothed with a Gaussian filter with $\sigma^2 = K_{smooth}L_{walls}$. At each time step the agent receives the visual input from the walls, determined by the intersection points of its visual field with the walls. When the agent reaches a distance of L_yL_{buffer} from the wall, it turns to the opposite direction.

Tuning properties of units

For each unit we calculated a tuning curve. We divided the arena into 100 equal bins and computed the number of time steps in each bin and the mean unit activation. We then obtained the tuning curve by dividing the mean activity for each bin by the occupancy. We treated movement in each direction as a separate location. We calculated the spatial information (SI) of the tuning curves for each unit:

$$SI = \sum_i p_i \frac{r_i}{\bar{r}} \log_2 \frac{r_i}{\bar{r}} \quad (4)$$

where i is the index of the bin, p_i is the probability of being in the bin, r_i is the value of the tuning curve in the bin and $\bar{r}$ is the unit's mean activity rate. Active unit was defined as a unit with non-zero activation for at least one input.

Simulations

For the random simulations, we train each network for 10^7 training steps while choosing random learning algorithm and parameters. The ranges and relevant values of parameters are specified in **Table 1** [↗](#). For Adam and SED there was no added noise.

Stochastic Error Descent

The equation for parameter updates under this learning rule is given by:

$$\theta_{\tau+1} = \theta_{\tau} - \eta(f(\theta_{\tau} + \xi_{\tau}) - f(\theta_{\tau}))\xi_{\tau} \quad (5)$$

In this learning rule, the parameters are randomly perturbed at each training step by a Gaussian noise denoted by ξ_{τ} and then updated in proportion to the change in loss.

Label noise

Label noise is introduced to the loss function given by the following formula:

$$f(\mathbf{x}_t) = (\hat{\mathbf{y}}_t - \mathbf{x}_{t+1} + \xi_{\tau}^{label})^2, \quad (6)$$

where ξ_{τ}^{label} is Gaussian noise.

Gradient descent dynamics around the zero-loss manifold

The function we used for the two-dimensional example was given by:

$$L(x, y) = (xy)^2, \quad (7)$$

which has zero loss on the x and y axes. For small enough update noise, GD will converge to the vicinity of this manifold (the axes). We consider a point on the x axis: $(x_0, 0)$, and calculate the direction of the gradient near that point. Because we are interested in motion along the zero-loss manifold, we consider a small perturbation in the orthogonal direction $(x_0, 0 + \Delta y)$ where $x_0 \gg 1$ and $|\Delta y| \ll 1$. Any component of the gradient in the x direction will lead to motion along the manifold. The update step at this point is given by:

$$-\nabla L(x_0, 0 + \Delta y) = -2x_0 \begin{pmatrix} (\Delta y)^2 \\ x_0 \Delta y \end{pmatrix}. \quad (8)$$

One can observe that the step has a large component in the y direction, quickly returning to the manifold. There is also a smaller component in the x direction, reducing the value of x . Reducing x also reduces the Hessian's eigenvalues:

$$H_L(x_0, 0) = 2 \begin{pmatrix} 0 & 0 \\ 0 & x_0^2 \end{pmatrix} \quad (9)$$

$$\lambda_{1,2} = \{0, x_0^2\}, v_{1,2} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}. \quad (10)$$

Thus, it becomes clear that the trajectory will have a bias that reduces the curvature in the y direction.

For general loss functions and various noise models, rigorous proofs can be found in [25], and a different approach can be found in [26]. Here, we will briefly outline the intuition for the general case. Consider again the update rule for GD:

$$\theta \leftarrow \theta - \eta \nabla L(\theta). \quad (11)$$

In order to understand the dynamics close to the zero-loss manifold, we consider a point θ , for which $L(\theta) = 0$ expand the loss around it:

$$L(\theta + \delta\theta) = L(\theta) + \nabla^T L(\theta) \delta\theta + \frac{1}{2} \delta\theta^T H \delta\theta. \quad (12)$$

We can then take the gradient of this expansion with respect to θ :

$$\nabla_{\theta} L(\theta + \delta\theta) = \nabla_{\theta} L(\theta) + \nabla_{\theta} \nabla_{\theta}^T L(\theta) \delta\theta + \nabla_{\theta} \left(\frac{1}{2} \delta\theta^T H \delta\theta \right) \quad (13)$$

$$= 0 + H \delta\theta + \nabla_{\theta} \left(\frac{1}{2} \delta\theta^T H \delta\theta \right). \quad (14)$$

The first term is zero, because the gradient is zero on the manifold. The second term is the largest one, as it linear in $\delta\theta$. Note that the Hessian matrix has zero eigenvalues in directions on the zero-loss manifold, and non-zero eigenvalues in other directions. Thus, the second term corresponds to projecting $\delta\theta$ in a direction that is orthogonal to the zero-loss manifold. The third term can be interpreted as the gradient of some auxiliary loss function. Thus, we expect gradient descent to minimize this new loss, which corresponds to a quadratic form with the Hessian. This is the reason for the implicit regularization along the manifold. Note that the auxiliary loss function is defined by $\delta\theta$, and thus different noise statistics will correspond, on average, to different implicit regularizations. In conclusion, the update step will have a large component that moves the parameter vector towards the zero-loss manifold, and a small component that moves the parameter vector on the manifold in a direction that minimizes some measure of the Hessian.

Hessian and sparseness

In the main text, we show that the implicit regularization of the Hessian leads to sparse representations. Here, we show this relationship for a single-hidden layer feed-forward neural network with ReLU activation and Mean Squared Error loss:

$$f(\mathbf{x}_i) = \sigma(\mathbf{x}_i \mathbf{m}^T + b) \mathbf{n}^T \quad (15)$$

The gradient and Hessian at the zero-loss manifold are given by [45]:

$$\nabla_{\theta} f(\mathbf{x}_i) = \begin{pmatrix} \frac{\partial f}{\partial \mathbf{m}} \\ \frac{\partial f}{\partial b} \\ \frac{\partial f}{\partial \mathbf{n}} \end{pmatrix} = \begin{pmatrix} \mathbf{n} \odot \mathbb{I}(\mathbf{x}_i; \theta) \odot \mathbf{x}_i \\ \mathbf{n} \odot \mathbb{I}(\mathbf{x}_i; \theta) \\ (\mathbf{x}_i \cdot \mathbf{n}^T + b) \odot \mathbb{I}(\mathbf{x}_i; \theta) \end{pmatrix} \quad (16)$$

$$\nabla_{\theta}^2 L(\mathbf{x}; \theta) = \sum_i \nabla_{\theta} f(\mathbf{x}_i) \nabla_{\theta} f(\mathbf{x}_i)^T, \quad (17)$$

where $\mathbb{I}(\mathbf{x}_i; \theta)$ is an indicator vector denoting whether each unit is active for some input $\mathbf{x}_i$. Sparseness means that a unit has become inactive for all inputs. All the partial derivatives of input, output and bias weights associated with such a unit are zero, and thus the relevant rows of the Hessian are zero as well. Thus, every inactive unit leads to several zero eigenvalues.

Acknowledgements

We thank Ron Teichner and Kabir Dabholkar for comments on the manuscript. This research was supported by the ISRAEL SCIENCE FOUNDATION (grants Nos. 2655/18 and 2183/21 to DD, and 1442/21 to OB), by the German-Israeli Foundation (GIF I-1477-421.13/2018) to DD, by a grant from the US-Israel Binational Science Foundation (NIMH-BSF CRCNS BSF:2019807, NIMH:R01 MH125544-01 to DD), by an HFSP research grant (RGP0017/2021) to OB, A Rappaport Institute Collaborative research grant to DD, by Israel PBC-VATAT and by the Technion Center for Machine Learning and Intelligent Systems (MLIS) to DD and OB, by the Prince Center for the Aging Brain, and by a University of Michigan – Israel Partnership for Research and Education Collaborative Research stipend to DK. (data science)

References

- [1] O'Keefe John, Nadel Lynn (1979) **The hippocampus as a cognitive map** *Behavioral and Brain Sciences* **2**:487–494
- [2] O'Keefe John, Dostrovsky Jonathan (1971) **The hippocampus as a spatial map: preliminary evidence from unit activity in the freely-moving rat** *Brain research*
- [3] Hubel David H, Wiesel Torsten N (1962) **Receptive fields, binocular interaction and functional architecture in the cat's visual cortex** *The Journal of physiology* **160**
- [4] McNaughton Bruce L, Mizumori SJY, Barnes CA, Leonard BJ, Marquis M, Green EJ (1994) **Cortical representation of motion during unrestrained spatial navigation in the rat** *Cerebral Cortex* **4**:27–39
- [5] Ziv Yaniv, Burns Laurie D, Cocker Eric D, Hamel Elizabeth O, Ghosh Kunal K, Kitch Lacey J, El Gamal Abbas, Schnitzer Mark J (2013) **Long-term dynamics of ca1 hippocampal place codes** *Nature neuroscience* **16**:264–266
- [6] Driscoll Laura N., Pettit Noah L., Minderer Matthias, Chettih Selmaan N., Harvey Christopher D. (2017) **Dynamic Reorganization of Neuronal Activity Patterns in Parietal Cortex** *Cell* **170**:986–999
- [7] Deitch Daniel, Rubin Alon, Ziv Yaniv (2021) **Representational drift in the mouse visual cortex** *Current biology* **31**:4327–4339
- [8] Schoonover Carl E, Ohashi Sarah N, Axel Richard, Fink Andrew J P (2021) **Representational drift in primary olfactory cortex** *Nature* **594**
- [9] Liberti William A, Schmid Tobias A, Forli Angelo, Snyder Madeleine, Yartsev Michael M (2022) **Publisher correction: A stable hippocampal code in freely flying bats** *Nature* **606**
- [10] Sadeh Sadra, Clopath Claudia (2022) **Contribution of behavioural variability to representational drift** *Elife* **11**
- [11] Rule Michael E, O'Leary Timothy, Harvey Christopher D (2019) **Causes and consequences of representational drift** *Curr. Opin. Neurobiol* **58**:141–147
- [12] Driscoll Laura N, Duncker Lea, Harvey Christopher D (2022) **Representational drift: Emerging theories for continual learning and experimental future directions** *Current Opinion in Neurobiology* **76**
- [13] Ziv Noam E, Brenner Naama (2018) **Synaptic tenacity or lack thereof: spontaneous remodeling of synapses** *Trends in neurosciences* **41**:89–99
- [14] Rubin Alon, Geva Nitzan, Sheintuch Liron, Ziv Yaniv (2015) **Hippocampal ensemble dynamics timestamp events in long-term memory** *elife* **4**
- [15] Aitken Kyle, Garrett Marina, Olsen Shawn, Mihalas Stefan (2022) **The geometry of representational drift in natural and artificial neural networks** *PLOS Computational Biology* **18**

- [16] Kappel David, Habenschuss Stefan, Legenstein Robert, Maass Wolfgang (2015) **Network plasticity as bayesian inference** *PLoS computational biology* **11**
- [17] Rokni Uri, Richardson Andrew G, Bizzi Emilio, Sebastian Seung H (2007) **Motor learning with unstable neural representations** *Neuron* **54**:653–666
- [18] Susman Lee, Brenner Naama, Barak Omri (2019) **Stable memory with unstable synapses** *Nature communications* **10**
- [19] Mongillo Gianluigi, Rumpel Simon, Loewenstein Yonatan (2017) **Intrinsic volatility of synaptic connections—a challenge to the synaptic trace theory of memory** *Current opinion in neurobiology* **46**:7–13
- [20] Kossio Yaroslav Felipe Kalle, Goedeke Sven, Klos Christian, Memmesheimer Raoul-Martin (2021) **Drifting assemblies for persistent memory: Neuron transitions and unsupervised compensation** *Proceedings of the National Academy of Sciences* **118**
- [21] Khatib Dorgham, Ratzon Aviv, Sellevoll Mariell, Barak Omri, Morris Genela, Derdikman Dori (2023) **Active experience, not time, determines within-day representational drift in dorsal ca1** *Neuron*
- [22] Geva Nitzan, Deitch Daniel, Rubin Alon, Ziv Yaniv (2023) **Time and experience differentially affect distinct aspects of hippocampal representational drift** *Neuron*
- [23] Qin Shanshan, Farashahi Shiva, Lipshutz David, Sengupta Anirvan M, Chklovskii Dmitri B, Pehlevan Cengiz (2023) **Coordinated drift of receptive fields in hebbian/anti-hebbian network models during noisy representation learning** *Nature Neuroscience* :1–11
- [24] Pashakhanloo Farhad, Koulakov Alexei (2023) **Stochastic gradient descent-induced drift of representation in a two-layer neural network** *arXiv preprint*
- [25] Blanc Guy, Gupta Neha, Valiant Gregory, Valiant Paul (2020) **Implicit regularization for deep neural networks driven by an ornstein-uhlenbeck like process** *In Conference on learning theory* :483–513
- [26] Li Zhiyuan, Wang Tianhao, Arora Sanjeev (2021) **What happens after sgd reaches zero loss?—a mathematical framework** *arXiv preprint*
- [27] Bengio Yoshua, Lee Dong-Hyun, Bornschein Jorg, Mesnard Thomas, Lin Zhouhan (2015) **Towards biologically plausible deep learning** *arXiv preprint*
- [28] Liu Yuhua Helena, Ghosh Arna, Richards Blake, Shea-Brown Eric, Lajoie Guillaume (2022) **Beyond accuracy: generalization properties of bio-plausible temporal credit assignment rules** *Advances in Neural Information Processing Systems* **35**:23077–23097
- [29] Marschall Owen, Cho Kyunghyun, Savin Cristina (2020) **A unified framework of online learning algorithms for training recurrent neural networks** *The Journal of Machine Learning Research* **21**:5320–5353
- [30] Recanatesi Stefano, Farrell Matthew, Lajoie Guillaume, Deneve Sophie, Rigotti Mattia, Shea-Brown Eric (2021) **Predictive learning as a network mechanism for extracting low-dimensional latent space representations** *Nature Communications* **12**

- [31] Karlsson Mattias P, Frank Loren M (2008) **Network dynamics underlying the formation of sparse, informative representations in the hippocampus** *Journal of Neuroscience* **28**:14271–14281
- [32] Sheintuch Liron, Rubin Alon, Ziv Yaniv (2022) **Bias-free estimation of information content in temporally sparse neuronal activity** *PLoS computational biology* **18**
- [33] Kingma Diederik P, Ba Jimmy (2014) **Adam: A method for stochastic optimization** *arXiv preprint*
- [34] Hinton Geoffrey, Srivastava Nitish, Swersky Kevin (2012) **Neural networks for machine learning lecture 6a overview of mini-batch gradient descent** *Cited on* **14**
- [35] Cauwenberghs Gert (1992) **A fast stochastic error-descent algorithm for supervised learning and optimization** *Advances in neural information processing systems* **5**
- [36] Mulayoff Rotem, Michaeli Tomer, Soudry Daniel (2021) **The implicit bias of minima stability: A view from function space** *Advances in Neural Information Processing Systems* **34**:17749–17761
- [37] Nguyen Nghia D, Lutas Andrew, Fernando Jesseba, Vergara Josselyn, McMahon Justin, Dimidschstein Jordane, Andermann Mark L (2022) **Cortical reactivations predict future sensory responses** *bioRxiv* :2022–11
- [38] Brette Romain (2019) **Is coding a relevant metaphor for the brain?** *Behavioral and Brain Sciences* **42**
- [39] Turner Elia, Dabholkar Kabir V, Barak Omri (2021) **Charting and navigating the space of solutions for recurrent neural networks** *Advances in Neural Information Processing Systems* **34**:25320–25333
- [40] Sanders Honi, Wilson Matthew A, Gershman Samuel J (2020) **Hippocampal remapping as hidden state inference** *Elife* **9**
- [41] Richards Blake A, Lillicrap Timothy P, Beaudoin Philippe, Bengio Yoshua, Bogacz Rafal, Christensen Amelia, Clopath Claudia, Costa Rui Ponte, de Berker Archy, Ganguli Surya, et al. (2019) **A deep learning framework for neuroscience** *Nature neuroscience* **22**:1761–1770
- [42] Marblestone Adam H, Wayne Greg, Kording Konrad P (2016) **Toward an integration of deep learning and neuroscience** *Frontiers in computational neuroscience*
- [43] Saxe Andrew, Nelli Stephanie, Summerfield Christopher (2021) **If deep learning is the answer, what is the question?** *Nature Reviews Neuroscience* **22**:55–67
- [44] Codoñer Francisco M, Darós José-Antonio, Solé Ricard V, Elena Santiago F (2006) **The fittest versus the flattest: experimental confirmation of the quasispecies effect with subviral pathogens** *PLoS pathogens* **2**
- [45] Nacson Mor Shpigel, Mulayoff Rotem, Ongie Greg, Michaeli Tomer, Soudry Daniel (2023) **The implicit bias of minima stability in multivariate shallow reLU networks** *inproceedings*

Article and author information

Aviv Ratzon

Rappaport Faculty of Medicine, Technion - Israel Institute of Technology, Haifa 31096, Israel,
Network Biology Research Laboratory, Technion - Israel Institute of Technology, Haifa 32000,
Israel

For correspondence: Aviv.Ratzon@hotmail.com

ORCID iD: [0009-0000-7648-9744](https://orcid.org/0009-0000-7648-9744)

Dori Derdikman

Rappaport Faculty of Medicine, Technion - Israel Institute of Technology, Haifa 31096, Israel
ORCID iD: [0000-0003-3677-6321](https://orcid.org/0000-0003-3677-6321)

Omri Barak

Rappaport Faculty of Medicine, Technion - Israel Institute of Technology, Haifa 31096, Israel,
Network Biology Research Laboratory, Technion - Israel Institute of Technology, Haifa 32000,
Israel
ORCID iD: [0000-0002-7894-6344](https://orcid.org/0000-0002-7894-6344)

Copyright

© 2023, Ratzon et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/) (<https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Adrien Peyrache

McGill University, Canada

Senior Editor

Laura Colgin

University of Texas at Austin, United States of America

Reviewer #1 (Public Review):

The authors start from the premise that neural circuits exhibit "representational drift" -- i.e., slow and spontaneous changes in neural tuning despite constant network performance. While the extent to which biological systems exhibit drift is an active area of study and debate (as the authors acknowledge), there is enough interest in this topic to justify the development of theoretical models of drift.

The contribution of this paper is to claim that drift can reflect a mixture of "directed random motion" as well as "steady state null drift." Thus far, most work within the computational neuroscience literature has focused on the latter. That is, drift is often viewed to be a harmless byproduct of continual learning under noise. In this view, drift does not affect the performance of the circuit nor does it change the nature of the network's solution or representation of the environment. The authors aim to challenge the latter viewpoint by showing that the statistics of neural representations can change (e.g. increase in sparsity)

during early stages of drift. Further, they interpret this directed form of drift as "implicit regularization" on the network.

The evidence presented in favor of these claims is concise. Nevertheless, on balance, I find their evidence persuasive on a theoretical level -- i.e., I am convinced that implicit regularization of noisy learning rules is a feature of most artificial network models. This paper does not seem to make strong claims about real biological systems. The authors do cite circumstantial experimental evidence in line with the expectations of their model (Khatib et al. 2022), but those experimental data are not carefully and quantitatively related to the authors' model.

To establish the possibility of implicit regularization in artificial networks, the authors cite convincing work from the machine-learning community (Blanc et al. 2020, Li et al., 2021). Here the authors make an important contribution by translating these findings into more biologically plausible models and showing that their core assumptions remain plausible. The authors also develop helpful intuition in Figure 4 by showing a minimal model that captures the essence of their result.

In Figure 2, the authors show a convincing example of the gradual sparsification of tuning curves during the early stages of drift in a model of 1D navigation. However, the evidence presented in Figure 3 could be improved. In particular, 3A shows a histogram displaying the fraction of active units over 1117 simulations. Although there is a spike near zero, a sizeable portion of simulations have greater than 60% active units at the end of the training, and critically the authors do not characterize the time course of the active fraction for every network, so it is difficult to evaluate their claim that "all [networks] demonstrated... [a] phase of directed random motion with the low-loss space." It would be useful to revise the manuscript to unpack these results more carefully. For example, a histogram of $\log(\tau)$ computed in panel B on a subset of simulations may be more informative than the current histogram in panel A.

Reviewer #2 (Public Review):

Summary:

In the manuscript "Representational drift as a result of implicit regularization" the authors study the phenomenon of representational drift (RD) in the context of an artificial network that is trained in a predictive coding framework. When trained on a task for spatial navigation on a linear track, they found that a stochastic gradient descent algorithm led to a fast initial convergence to spatially tuned units, but then to a second very slow, yet directed drift which sparsified the representation while increasing the spatial information. They finally show that this separation of timescales is a robust phenomenon and occurs for a number of distinct learning rules.

Strengths:

This is a very clearly written and insightful paper, and I think people in the community will benefit from understanding how RD can emerge in such artificial networks. The mechanism underlying RD in these models is clearly laid out and the explanation given is convincing.

Weaknesses:

It is unclear how this mechanism may account for the learning of multiple environments. The process of RD through this mechanism also appears highly non-stationary, in contrast to what is seen in familiar environments in the hippocampus, for example.

Reviewer #3 (Public Review):

Summary:

Single-unit neural activity tuned to environmental or behavioral variables gradually changes over time. This phenomenon, called representational drift, occurs even when all external variables remain constant, and challenges the idea that stable neural activity supports the performance of well-learned behaviors. While a number of studies have described representational drift across multiple brain regions, our understanding of the underlying mechanism driving drift is limited. Ratzon et al. propose that implicit regularization - which occurs when machine learning networks continue to reconfigure after reaching an optimal solution - could provide insights into why and how drift occurs in neurons. To test this theory, Ratzon et al. trained a Feedforward Network to perform the oft-utilized linear track behavioral paradigm and compare the changes in hidden layer units to those observed in hippocampal place cells recorded in awake, behaving animals.

Ratzon et al. clearly demonstrate that hidden layer units in their model undergo consistent changes even after the task is well-learned, mirroring representational drift observed in real hippocampal neurons. They show that the drift occurs across three separate measures: the active proportion of units (referred to as sparsification), spatial information of units, and correlation of spatial activity. They continue to address the conditions and parameters under which drift occurs in their model to assess the generalizability of their findings. However, the generalizability results are presented primarily in written form: additional figures are warranted to aid in reproducibility. Last, they investigate the mechanism through which sparsification occurs, showing that the flatness of the manifold near the solution can influence how the network reconfigures. The authors suggest that their findings indicate a three-stage learning process: 1) fast initial learning followed by 2) directed motion along a manifold which transitions to 3) undirected motion along a manifold.

Overall, the authors' results support the main conclusion that implicit regularization in machine learning networks mirrors representational drift observed in hippocampal place cells. However, additional figures/analyses are needed to clearly demonstrate how different parameters used in their model qualitatively and quantitatively influence drift. Finally, the authors need to clearly identify how their data supports the three-stage learning model they suggest. Their findings promise to open new fields of inquiry into the connection between machine learning and representational drift and generate testable predictions for neural data.

Strengths:

1. Ratzon et al. make an insightful connection between well-known phenomena in two separate fields: implicit regularization in machine learning and representational drift in the brain. They demonstrate that changes in a Feedforward Network mirror those observed in the brain, which opens a number of interesting questions for future investigation.
2. The authors do an admirable job of writing to a large audience and make efforts to provide examples to make machine learning ideas accessible to a neuroscience audience and vice versa. This is no small feat and aids in broadening the impact of their work.

3. This paper promises to generate testable hypotheses to examine in real neural data, e.g., that drift rate should plateau over long timescales (now testable with the ability to track single-unit neural activity across long time scales with calcium imaging and flexible silicon probes). Additionally, it provides another set of tools for the neuroscience community at large to use when analyzing the increasingly high-dimensional data sets collected today.

Weaknesses:

1. Neural representational drift and directed/undirected random walks along a manifold in ML are well described. However, outside of the first section of the main text, the analysis focuses primarily on the connection between manifold exploration and sparsification without addressing the other two drift metrics: spatial information and place field correlations. It is therefore unclear if the results from Figures 3 and 4 are specific to sparseness or extend to the other two metrics. For example, are these other metrics of drift also insensitive to most of the parameters as shown in Figure 3 and the related text? These concerns could be addressed with panels analogous to Figures 3a-c and 4b for the other metrics and will increase the reproducibility of this work.
2. Many caveats/exceptions to the generality of findings are mentioned only in the main text without any supporting figures, e.g., "For label noise, the dynamics were qualitatively different, the fraction of active units did not reduce, but the activity of the units did sparsify" (lines 116-117). Supporting figures are warranted to illustrate which findings are "qualitatively different" from the main model, which are not different from the main model, and which of the many parameters mentioned are important for reproducing the findings.
3. Key details of the model used by the authors are not listed in the methods. While they are mentioned in reference 30 (Recanatesi et al., 2021), they need to be explicitly defined in the methods section to ensure future reproducibility.
4. How different states of drift correspond to the three learning stages outlined by the authors is unclear. Specifically, it is not clear where the second stage ends, and the third stage begins, either in real neural data or in the figures. This is compounded by the fact that the third stage - of undirected, random manifold exploration - is only discussed in relation to the introductory Figure 1 and is never connected to the neural network data or actual brain data presented by the authors. Are both stages meant to represent drift? Or is only the second stage meant to mirror drift, while undirected random motion along a manifold is a prediction that could be tested in real neural data? Identifying where each stage occurs in Figures 2C and E, for example, would clearly illustrate which attributes of drift in hidden layer neurons and real hippocampal neurons correspond to each stage.